$0.1 < \lambda_{TARGET} < 0.15$ $\quad R_w = 0.2\ m$

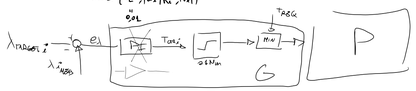
0.06

$$\lambda_i = \frac{\omega_i R_w - V_x}{\max(\varepsilon, |\omega_i R_w|, |V_x|)}$$

oppure $\lambda_i = \dfrac{\omega_i R - v_{xi}}{\max(\varepsilon, v_{xi})}$

$$v_{xi} = \begin{cases} v_{xRL} = v_x - \psi\frac{w}{2} \\ v_{xRR} = v_x + \psi\frac{w}{2} \end{cases}$$

$e\lambda \begin{cases} \geq 0 & \text{GRIP OK} \to \text{uscito} = 0 \\ < 0 & \to \text{TORQUE} \\ \lambda_{MEAS} > \lambda_{TARGET} \end{cases}$



$T_{FINAL} = T_{ORQ_TV} - T_{ORQ}$

$$\dot{\omega}_{iL} = \frac{T_{m,i} - (F_{xi} \cdot R_w)}{J_w}$$

RUOTA L

$T_{m,i}$: COPPIA MOTORE SIMMETRICO

F_{xi}: FORZA CONTROMANO SPOSTA

J_w: INERZIA RUOTA

$\dot{\omega}_L = R_w$ BRAC QUOTA

FORZA RESISTENTE

$\lambda = \frac{V_w - V}{V_w}$ $\quad w = \frac{V}{V(\lambda=1)}$

CONTROLLO IN VELOCITÀ

$F_{xL} \cdot R_w$ = COPPIA RESISTENTE

$T_{m,L} - F_{xL} \cdot R_w$ = COPPIA MOTORE

$\dot{\omega}_L$ = ACC. ANGOLARE RUOTA L

$\dot{\omega}_L \to \boxed{\frac{1}{s}} \to \omega_L$

condiz. $= 0$

21/04/26

$$V_x = \frac{\omega_{FL} + \omega_{FR}}{2} \cdot R_w$$

$$\lambda_i = \frac{\omega_{Ri} \cdot R_w - V_x}{\max(V_x, V_{x\,MIN})}$$

$i = RR, RL$ $\quad V_{x\,MIN} \approx 1$

MEGLIO × CURVA

$$\lambda_i = \frac{\omega_{Ri} R_w - V_{REF\,i}}{\max(V_{REF\,i}, V_{MIN})}$$

con $\begin{cases} V_{REF\,RL} = V_x - \psi\frac{w}{2} \\ V_{REF\,RR} = V_x + \psi\frac{w}{2} \end{cases}$

$T_{OUT\,i} = K_i \cdot T_{REF\,i}$

- $\lambda > 0$ SCIVOLAMENTO (trazione + radice reali)
- $\lambda = 0$ rotolamento puro

$\lambda_{REF} \approx 0.1 \div 0.15$

$e_i = \lambda_{REF} - \lambda_i$

$\lambda_i > \lambda_{REF} \to e_i < 0 \Rightarrow$ PI riduce K_i

$K_i \in [0,1]$

$K_i = K_P e_i + K_I \int e_i dt \Rightarrow K_i \in [0,1]$

- $V_x < V_{low}$ (5 m/s) $\quad K_i = 1$
- $\lambda_i < \lambda_{ref} - \text{deadband}$ $\quad K_i = 1$
- $\lambda_i \geq \lambda_{REF}$ $\quad K_i < 1$

posso usare media velocità ruote fredde

$\omega_{FL} = \frac{V_x - \psi\frac{w}{2}}{R_w}$ $\quad \omega_{FR} = \frac{V_x + \psi\frac{w}{2}}{R_w}$

$\frac{\omega_{FL} + \omega_{FR}}{2} = \frac{V_x}{R_w}$

	INPUT			OUTPUT		
RUOTA POSTERIORE	$T_{RL\,REF}$	$T_{RR\,REF}$	[Nm]	$T_{RL\,OUT}$	T_{ROUT}	[Nm]
ENCODER	ω_{FL}	ω_{FR}	[rad/s]			
IMU	ω_{RL}	ω_{RR}	[rad/s]			
	$\dot{\psi}$					

× VEHICLE SPEED ESTIMATOR $\quad \omega_{FL}, \omega_{FR} \Rightarrow V_x$

$V_x \to$ ENABLE $\quad (V_x > V_{low})$

× SLIP ESTIMATOR R_L $\quad (V_{REF\,RL})$

$\omega_{RL}, V_x \to \lambda_{RL}$

× SLIP ESTIMATOR R_R $\quad (V_{REF\,RR})$

$\omega_{RR}, V_x \to \lambda_{RR}$

SATURAZIONE

× TC PI R_L $\quad \lambda_{RL}$, ENABLE $\to K_{RL}$ $\quad \in [0,1]$

× TC PI R_R $\quad \lambda_{RR}$, ENABLE $\to K_{RR}$ $\quad \in [0,1]$

$T_{RL\,OUT} = \lambda_{RL} \cdot T_{RL\,REF}$

$T_{RR\,OUT} = \lambda_{RR} \cdot T_{RR\,REF}$

× VEHICLE SPEED ESTIMATOR

$\frac{\omega_{FL} + \omega_{FR}}{2} \cdot R_w = V_x$

$V_{x\,SAFE} = \max(V_x, V_{x\,MIN})$

(max) ENABLE $= V_x > V_{low}$

× SLIP ESTIMATOR i $\quad V_{REF\,i} = V_{x\,SAFE} \pm \psi\frac{w}{2}$ $\quad + RR$, $- RL$

$$\frac{\omega_{Ri} \cdot R_w - V_{REF\,i}}{V_{REF\,i}} = \lambda_i$$

× TC PI i $\quad$ usare DISCRETE TIME INTEGRATOR + ANTIWINDUP

$\lambda_{REF} - \lambda_i = e_i$

ANTI-WINDUP $\quad$ PARAM. CALC.

$PI(e_i) = K_{RAW}$

ENABLE $\begin{cases} 1 \to K_{RAW} \\ 0 \to 1 \end{cases} = K_i$

(SPEGNO COPPIA DIVERSA)

$T_{OUT\,i} = K_i \cdot T_{REF\,i}$

$R_w = 0.225\ m$

$\lambda_{REF} = 0.10$

deadband $= 0.02$

$V_{low} = 1.5\ m/s$

$V_{x\,MIN} = 0.5\ m/s$

$w = 1.2\ m$